

Lecture 27

Properties of Fourier Transform

Property 0

Linearity of Fourier Transform

$$\text{If } \begin{array}{ccc} x & \xleftrightarrow{F} & X \\ y & \xleftrightarrow{F} & Y \end{array}$$

$$\alpha x + \beta y \xleftrightarrow{F} \alpha X + \beta Y$$

$$\alpha, \beta \in \mathbb{C}$$

Proof simply uses

$$\int_{-\infty}^{\infty} (\alpha x(t) + \beta y(t)) e^{-j\omega t} dt$$

↓

$$\alpha \int x(t) \cdot e^{-j\omega t} dt$$

$$+ \beta \int y(t) \cdot e^{-j\omega t} dt.$$

(1) Time shift

$$x(t) \xleftrightarrow{F} X(\omega)$$

and $t_0 \in \mathbb{R}$

$$x(t - t_0) \xleftrightarrow{F} ??$$

$$\begin{aligned} & \int_{-\infty}^{\infty} x(t - t_0) \cdot e^{-j\omega t} dt \\ &= \int_{-\infty}^{\infty} x(\tau) e^{-j\omega(\tau + t_0)} d\tau \end{aligned}$$

$$\tau = t - t_0$$

$$= \int_{-\infty}^{+\infty} x(\tau) e^{-j\omega\tau} \cdot e^{-j\omega t_0} d\tau$$

$$= e^{-j\omega t_0} \int_{-\infty}^{+\infty} x(\tau) e^{-j\omega\tau} d\tau$$

$$= e^{-j\omega t_0} \cdot X(\omega).$$

$$x(t) \xleftrightarrow{F} X(\omega)$$

$$x(t-t_0) \xleftrightarrow{F} X(\omega) \cdot e^{-j\omega t_0}$$

$$\text{Note: } |X(\omega) e^{-j\omega t_0}| = |X(\omega)|$$

$$\angle X(\omega) \cdot e^{-j\omega t_0} = \angle X(\omega) - \omega t_0$$

2. Frequency Shifting

$$x(t) \xleftrightarrow{F} X(\omega)$$

choose any $\omega_0 \in \mathbb{R}$

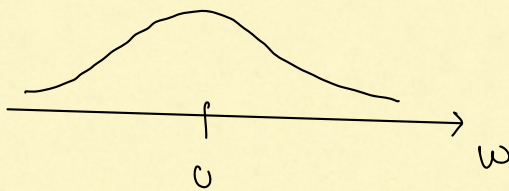
$$x(t) \cdot e^{j\omega_0 t} \xleftrightarrow{F} ??$$

$$\int_{-\infty}^{+\infty} x(t) \cdot e^{j\omega_0 t} \cdot e^{-j\omega t} dt = \int_{-\infty}^{+\infty} x(t) \cdot e^{-j(\omega - \omega_0)t} dt.$$

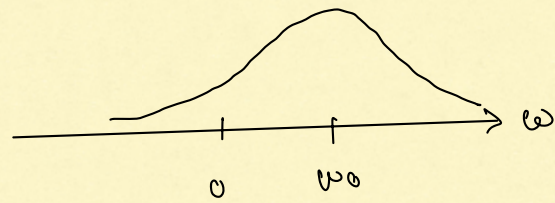
$$= X(\omega - \omega_0)$$

$$x(t) \cdot e^{j\omega_0 t} \xleftrightarrow{F} X(\omega - \omega_0).$$

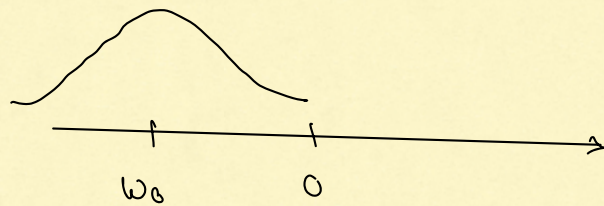
FT of $x(t)$



FT of $x(t) \cdot e^{j\omega_0 t}$, $\omega_0 > 0$



FT of $x(t) \cdot e^{j\omega_0 t}$ if $\omega_0 < 0$



3. Time scaling / Frequency scaling

$$x(t) \xleftrightarrow{F} X(\omega)$$

Pick any $x(at) \xleftrightarrow{F} ??$

$$a > 0$$

$$\int_{-\infty}^{+\infty} x(at) e^{-j\omega t} dt, \quad at = \tau$$

$$= \int_{-\infty}^{+\infty} x(\tau) \cdot e^{-j\omega \tau/a} \cdot \frac{d\tau}{a}$$

$$= \frac{1}{a} \int_{-\infty}^{+\infty} x(\tau) \cdot e^{-j(\frac{\omega}{a}) \tau} d\tau$$

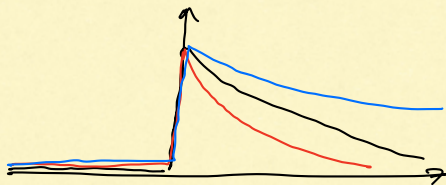
$$= \frac{1}{a} \cdot X\left(\frac{\omega}{a}\right).$$

If $a < 0$: $x(at) \xleftrightarrow{F} -\frac{1}{a} X\left(\frac{\omega}{a}\right)$.

For any $a \neq 0$, $x(at) \xleftrightarrow{F} \frac{1}{|a|} X\left(\frac{\omega}{a}\right)$.

$a \in \mathbb{R}$

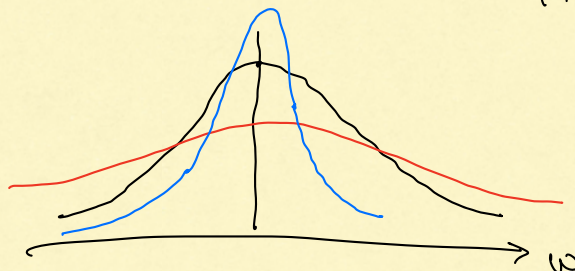
$e^{-t} u(t)$



$a > 1$

$a < 1$

$|X(\omega)| = \frac{1}{|1+j\omega|}$



$a = -1 \Rightarrow$ Time Reversal:

$x(-t) \xleftrightarrow{F} X(-\omega)$.

4. Conjugation

$x(t) \xleftrightarrow{F} X(\omega)$

$x^*(t) \xleftrightarrow{F} ??$

$$\begin{aligned} \int_{-\infty}^{+\infty} x^*(t) e^{-j\omega t} dt &= \left[\int_{-\infty}^{+\infty} x(t) \cdot e^{j\omega t} dt \right]^* \\ &= \left[\int_{-\infty}^{+\infty} x(t) \cdot e^{-j(-\omega)t} dt \right]^* \\ &= [X(-\omega)]^* = X^*(-\omega). \end{aligned}$$

$$x(t) \xleftrightarrow{F} X(\omega)$$

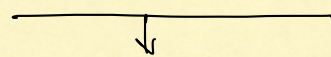
$$x^*(t) \xleftrightarrow{F} X^*(-\omega).$$

5. What if $x(t)$ is real-valued?

From property 4: $x(t) = x^*(t)$

$$\Rightarrow X(\omega) = X^*(-\omega).$$

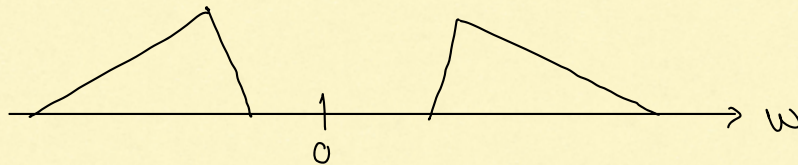
$$\Rightarrow X^*(\omega) = X(-\omega)$$



Hermitian symmetry or
Conjugate symmetry.

$$|X(-\omega)| = |X^*(\omega)| = |X(\omega)|.$$

$$|X(\omega)|$$



6. Parseval's Relation

$$\int_{-\infty}^{+\infty} |x(t)|^2 dt = \text{energy of a signal } x(t).$$

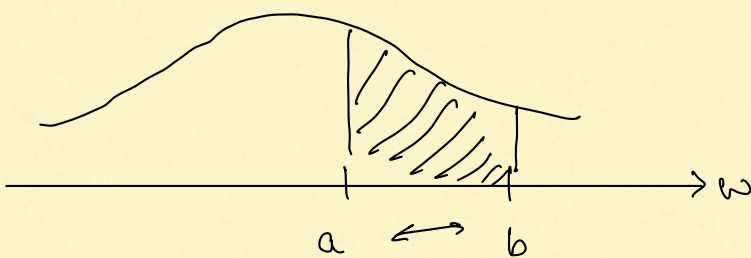
$$\int_{-\infty}^{+\infty} |x(t)|^2 dt = \frac{1}{2\pi} \int_{-\infty}^{+\infty} |X(\omega)|^2 d\omega.$$

$\frac{|X(\omega)|^2}{2\pi}$ = how the total energy is distributed across all frequencies.

$\int_a^b \frac{1}{2\pi} |X(\omega)|^2 d\omega$ = energy contained between the frequencies $a \leq \omega \leq b$.

= energy contained in the frequency band

$[a, b]$.



$\frac{|X(\omega)|^2}{2\pi}$ = energy density spectrum of $x(t)$.

Homework: Use Parseval's relation on $x(t) = \sin(t)$:

$$\int_{-\infty}^{+\infty} \sin^4(t) dt = ?$$

7. Multiplication Property.

Convolution Prop: If h, x are bounded and absolutely integrable

$$x * h \xleftrightarrow{F} X(\omega) \cdot H(\omega).$$

Under same mathematical conditions:

$$x(t) \cdot h(t) \xleftrightarrow{F} \frac{1}{2\pi} X(\omega) * H(\omega).$$

8. The Derivative Property

$$x(t) \xleftrightarrow{F} X(\omega)$$

If

x is absolutely integrable

$$x'(t) \xleftrightarrow{F} ??$$

x is differentiable at all time t

$$\underline{\omega = 0} : \int_{-\infty}^{+\infty} x'(t) \cdot e^{-j\omega t} dt \quad \text{at } \omega = 0$$

$$= \int_{-\infty}^{+\infty} x'(t) dt = [x(t)]_{-\infty}^{+\infty}$$

$$= x(+\infty) - x(-\infty). \quad \leftarrow x(+\infty) = 0, \quad x(-\infty) = 0.$$

$$= 0 - 0 = 0.$$

$$\underline{\omega \neq 0} : X(\omega) = \int_{-\infty}^{+\infty} x(t) \cdot e^{-j\omega t} dt$$

$$= \int_{-\infty}^{+\infty} x(t) d \left[\frac{e^{-j\omega t}}{-j\omega} \right]$$

$\begin{matrix} \longleftrightarrow & \longleftrightarrow \\ u & dv \end{matrix}$

$$= \left[x(t) \cdot \frac{e^{-j\omega t}}{-j\omega} \right]_{-\infty}^{+\infty} - \int_{-\infty}^{+\infty} x'(t) \frac{e^{-j\omega t}}{-j\omega} dt.$$

$$= [0 - 0] + \frac{1}{j\omega} \int_{-\infty}^{+\infty} x'(t) \cdot e^{-j\omega t} dt$$

$$\therefore \int_{-\infty}^{+\infty} x'(t) \cdot e^{-j\omega t} dt = j\omega \cdot X(\omega), \quad \omega \neq 0$$

Combining the cases $\omega = 0$ and $\omega \neq 0$:

$$\int_{-\infty}^{+\infty} x'(t) \cdot e^{-j\omega t} dt = j\omega \cdot X(\omega).$$

$$\therefore x'(t) \overset{F}{\longleftrightarrow} j\omega X(\omega).$$

9. Derivative of $X(\omega)$.

$$x(t) \overset{F}{\longleftrightarrow} X(\omega)$$

$$?? \overset{F}{\longleftrightarrow} X'(\omega)$$

If:

1. x is absolutely integrable
2. Fourier inversion theorem applies to x, X .
3. X is differentiable everywhere
4. $X(\omega) \rightarrow 0$ as $\omega \rightarrow +\infty$ and $\omega \rightarrow -\infty$.

$$-jt \cdot x(t) \overset{F}{\longleftrightarrow} X'(\omega).$$